

SUJAL

AI DEVELOPER

PYTHON | FASTAPI | MONGODB | BACKEND SYSTEMS

PROFESSIONAL SUMMARY

AI Developer with experience in Python, FastAPI, and MongoDB, building scalable systems. Solved 300+ LeetCode problems with strong DSA fundamentals. Developing LLM applications using RAG, MCP, and Groq APIs for real-world use.

CONTACT

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Portfolio: PORTFOLIO ↗

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, PHP, JavaScript(basic)

Backend & Frameworks: FastAPI, REST API development

Databases: SQL, PL/SQL, MongoDB, Vector DB

AI and ML: MCP, LLM Agents, RAG pipelines, Prompt Engineering, OpenAI API, TTS/STT integration

Data Structures & Algorithms: Strong foundation — 300+ LeetCode problems solved

Tools & Others: Git, GitHub, Postman, VS Code, AntiGravity, Linux

PROJECTS

LLM-Powered Movie Search Engine ↗

- Problem: Keyword search failed for natural language queries.
- Built a FastAPI backend using LLM-based query refinement with embeddings and vector similarity search for intent-aware retrieval.
- Improved relevance and discovery speed for complex queries like story-based searches.
- Stack: Python, FastAPI, MongoDB, Groq LLM, Embeddings, Similarity Search

Automated Email Engine using LLM and FastAPI ↗

- Problem: Manual email drafting was repetitive and time-consuming.
- Built a FastAPI service using an LLM to generate and send context-aware emails via SMTP.
- Reduced drafting effort and improved workflow efficiency through end-to-end AI automation.
- Stack: Python, FastAPI, LLM, SMTP, Streamlit

Real-time Speech-to-Speech AI ↗

- Problem: Text-based AI felt unnatural and disrupted conversation flow.
- Built a real-time voice pipeline (STT → LLM → TTS) for seamless voice-to-voice interaction with low latency.
- Enabled natural, hands-free AI conversations closer to human interaction.
- Stack: Python, STT, LLM, TTS, WebSockets, Streamlit

ACADEMIC HISTORY

Pursing BCA | PGGC-11 Chandigarh | 2023 to 2026

12th | HP Board Dharamshala | 2023 | 90.4%